Virtual University of Pakistan

BT405

Nano Biotechnology

Midterms Past Papers

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do it right if u found any mistake

solved by *Zareen Fatima* & Rania Ch

Bt405-mcqs

1. Nanotechnology_____ nanoscale
2. American Physicist name_____ Dr. Richard Feynman
3. By 2050, world population_____ _9.1billion
4. Human cell contain_____ 3 trillion
5. Gold properties wala-------
6. Diatom cell wall made of _____ silica (SiO2).
7. Nanotechnology not about _____ pm scale
8. Gold particles find significant exploitation_____ (i) their comparative chemical stability, making them less hazardous, (ii) simple and straightforward synthesis and fabrication process, and (iii) genuine biocompatibility and noninterference with other labeled biomaterials
9. Energy b/w valence n conduction band_____ Eg
10. Semiconductor have Eg_____ is between valence and conduction
11. Nanometer about_____10 H atom
12. Carbon tube width_____3nm

Nanotechnology

Ans: Simplest _ technology at the nonscale. Nanotechnology literally means any technology on a nanoscale that has applications in the real world. Succinct definition of nanotechnology is engineering with atomic precision or atomically precise technology _ APT. US National nanotechnology _ the essence of nanotechnology is the ability to work at the molecular level, atom-by-atom, to create large structure with fundamentally new molecular organization. Nanotechnology is group of emerging technology in which the structure of matter is controlled at the nanometer scale to produce novel materials and device that have useful and unique properties

What are Self Assembled Monolayers

Ans: Self-assembled monolayers (SAMs)

Some organic molecules, when exposed from solution or vapour to a suitable substrate, self-assemble to produce homogeneous, densely packed layers of monomolecular thickness. These organic molecules have long chains with two different end groups. monolayer is formed when one of the two end groups of the organic molecule reacts with a particular surface forming a chemical bond. The surface properties of the substrate are then defined by the exposed functional groups of the monolayer.

For example, alkyl-silane or alkane thiol molecules exposed to a silica or metal surface assemble into organised layers.

Environmental application of Metal Nano Particles

Ans: Some forms of metal nanoparticles have important environment applications. Zero-valent (Fe0) iron nanoparticles are under investigation for the remediation of contaminated groundwater and soil. Iron, when exposed to air, oxidises easily to rust; however, when it oxidises around contaminants such as trichloroethylene (TCE), carbon tetrachloride, dioxins, or PCBs, these organic molecules are broken down into simple, far less toxic carbon compounds.Iron nanoparticles are more effective than conventional 'iron powder', which is already used to clean up industrial wastes. Iron nanoparticles are 10 to 1 000 times more reactive than commonly used iron powders. Silver nanoparticles have a strong antibacterial capacity. They are used in numerous products to prevent or reduce the adherence of bacteria to surfaces

Applications of Nanotechnology in diagnostics

Ans: Nanodiagnostics is an evolving application of nanoscale technology to meet the demand of clinical diagnostics, determining disease state, any predisposition to such, the pathology of the condition and the identification of causative organisms. With nanotechnology, diagnosis is being carried out on a nano-scale leading to a trend of the use of hand held devices that are easy to use and marketable. Nanodiagnostics as surging new field of molecular diagnostics, have been positively changing laboratory procedures, providing new ways for patient's sample assessment and early detection of disease biomarkers with increased sensitivity and specificity. Nanoparticle platforms have been developed and optimized for the detection of pathogens and cancer biomarkers such that diagnostic proceedures now become less cumbersome but more sensitive because most of the complex proceedures are now integrated onto a simple device having the capacity to be used for on the spot diagnosis

Diatoms wala

Applications of Nanotechnology in the field of Medicin

Ans: **Diagnostic Applications of Nanotechnology in Medicine**

- Improved imaging of the human (or any) body
- Nanoprobes (miniature machines) can attach themselves to particles in the body (e.g., antibodies) and emit a magnetic field.
- Probes that aren't attached to anything don't create a detectable magnetic
- Nano-tracking may be able to detect tumors that are a few cells in size.
- A single inhaled nanorobot reaches, deeply inspired into the lungs, enters an alveolar duct and attaches to the tissue surface.
- Another way to use nanotech as tracking devices is to use "quantum dots
- These tiny semiconductors are able to emit wavelengths of light (colors) that depend on their size. If quantum dot A is twice as big as quantum dot B, it will emit a different color.
- Quantum dots are better than conventional dyes:

• They last much longer

• More colors can be made available.

A microscopic machine roaming through the bloodstream, injecting or taking samples for identification and determining the concentrations of different compounds

Therapeutic Applications of Nanotechnology in Medicine

- Nanotech is capable of delivering medication to the exact location where they are needed – hence lesser side effects

• Organic dendrimers - a type of artificial molecule roughly the size of a protein- would be ideal for the job of delivering medicine

• Hollow polymer capsules - gold-coated glass beads that are near infrared light sensitive

• Destruction of harmful eukaryotic organisms / cancer cells by interrupting their division process

• Certain proteins are capable of doing this (e.g., Bc12 family of proteins)

- Nanoprobe can be made to generate radiation, that could kill bacteria, viruses and cancer cells
- Nanoprobe comprising of a single caged actinium-225 atom would detect (using antibodies) and enter a cancerous cell
- Location and destruction of cancer cells by acoustic signals
- Comparison of normal and cancerous cells in respiratory airway of the lung
- Nanotechnology also theoretically allows the mimicking of natural biological processes e.g., repair of damaged tissues

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What is block copolymers (2)

Ans: A copolymer is a macromolecule containing two or more types of monomers and a block copolymer comprises these basic units or monomer types joined together in long individual sequences called blocks.
An example is the diblock polymer (A)m(B)n, which is made of a linear sequence of m monomers of type A joined together to a linear sequence of n monomers of type B.

Current uses of nanotechnology. (3)

Ans: Nanotechnology is not an industry; nor is it a single technology or a single field of research
The field of nanotechnology is interdisciplinary.
Nanotechnology is referred to as a general-purpose technology.
The total market for nanobiotechnology products was $19.3 billion in 2010 and is growing at a compound annual growth rate (CAGR) of 9% to reach a forecast market size of $29.7 billion by 2015.
Medical applications, including drug delivery and microbicides, dominate today's market, with sales of $19.1 billion in 2010. This market segment is growing at a compound annual growth rate (CAGR) of 8.7%, and is forecast to reach sales of $29 billion by 2015 (BCC).
More and more consumer products are branded worldwide as nano.

Explain self-healing adhesive and why diatoms are biommitic models of self healing materials. (5)

Ans:

- Diatoms are a type of algae with nanostructured amorphous silica surfaces.
- Diatoms are unicellular algae that have the capability to colonize any natural and manmade submerged surfaces.
- Diatoms are a large group of eukaryotic microalgae that have a rigid cell wall made of silica (SiO2).
- Some diatom species have evolved strong self-healing underwater adhesives. Some are free-floating, others have adhesive properties in water: for instance, diatoms in the Antarctic seas can attach to ice.

- Others secrete viscous mucilage which binds colonies together while protecting the silica shells from wear as they rub against each other.
- Molluscs are another species that have adhesive properties under water.
- Both diatoms and molluscs have strong underwater glues that can also resist stress and self-heal.
- They serve as a biomimetic model for self-healing materials.
- These natural adhesives and their self-healing properties are due to the properties of the proteins contained in them.
- These proteins have sacrificial bonds that allow the molecule to be reversibly stretched by breaking and re-bonding.
- This sacrificial bond behaviour has been observed in many other materials, such as wool.
- Closing the wounds with sticky materials has been introduced to prevent bleeding and induce the wound healing process.
- Many types of surgical adhesives including tissue adhesives have been developed to be a suitable alternative for sutures and staples.
- A new approach which aims at producing bio-adhesives by mimicking the nature along with applying nanotechnology methods.
- Engineering stable underwater adhesives currently poses a major technical challenge, because most man-made adhesives fail in wet conditions, owing to chemical modification of the adhesive or its substrate.
- Diatoms produce adhesives that are extremely strong and robust, in both fresh- and seawater environments.
- AFM (Atomic Force Microscopy) study of living diatoms, phase imaging and force spectroscopy experiments revealed characteristics of these natural adhesives that may be of use in designing man-made adhesive analogues that function in wet environments.
- The AFM allows for investigations of micromechanical properties of the cell surface, for example viscoeleastic properties, adhesion forces and hardness measurements, in physiological conditions.
- Experiments with three diatom species that produce outstanding natural adhesives: Eunotia sudetica, Navicula seminulum and an unidentified species

Three Applications of Biotechnology 2 marks

Ans: This reflects the current needs of humans- we have too many diseases and currently too few drugs to target them.

By far, cancer has the most drug candidates than any other disease. This disease has many more drug candidates than hepatitis C because hepatitis C affects less people worldwide than different kinds of cancer. Cancer, of course, can affect many different organs. On the other hand, hepatitis C only affects the liver.

Write the phases of crystal liquid 3 marks

Ans:

1. Nematic Phase: Molecules in this phase are long and rod-like in shape. They are free to move in space.
2. Smectic Phase: This phase can be reached at lower temperatures than the nematic phase. Molecules align themselves in layers. More order and higher viscosity.
3. Cholestric Liquid Crystal: Molecules with intermolecular forces that favor alignment between molecules at a slight angle to one another

Application of SAM 3 marks

Ans: Applications: SAMs are extremely useful in thin-film technology since they provide a simple, versatile and relatively inexpensive way of producing coatings with specific functionalities.

Thin coatings are used in many applications, for example to control surface wetting and adhesion properties; to provide chemical resistance or specific biocompatibility; to confer antibacterial properties; and for sensors.

SAMs are particularly interesting for developing DNA and protein micro and nanoarrays for medical screening and for future electronic components in computer chips and other ICT developments

ways nanotechnology impacts our lives daily basis?.) 5 marks

Ans: 1. Faster, smaller, and more powerful computers that consume far less power, with longerlasting batteries

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2. Faster, more functional, and more accurate medical diagnostic equipment. Lab-on-a-chip technology enables point-of-care testing in real time, which speeds up delivery of medical care. Nanomaterial surfaces on implants improve wear and resist infection.

3. Nanoparticles in pharmaceutical products improve their absorption within the body and make them easier to deliver, often through combination medical devices. Nanoparticles can also be used to deliver chemotherapy drugs to specific cells, such as cancer cells.

4. Improved vehicle fuel efficiency and corrosion resistance by building vehicle parts from nanocomposite materials that are lighter, stronger, and more chemically resistant than metal. Nanofilters remove nearly all airborne particles from the air before it reaches the combustion chamber, further improving gas mileage.

5. Nanoparticles or nanofibers in fabrics can enhance stain resistance, water resistance, and flame resistance, without a significant increase in weight, thickness, or stiffness of the fabric. For example, “nano-whiskers” on pants make them resistant to water and stains.

6. Water filters that are only 15-20 nanometers wide can remove nano-sized particles, including virtually all viruses and bacteria. These cost-efficient, portable water treatment systems are ideal for improving the quality of drinking water in emerging countries.

7. Carbon nanotubes have a variety of commercial uses, including making sports equipment stronger and lighter weight. For example, a tennis racket made with carbon nanotubes bends less during impact, and increases the force and accuracy of the delivery. Nanoparticle-treated tennis balls can keep bouncing twice as long as standard tennis balls.

8. Most sunscreens today are made from nanoparticles that effectively absorb light, including the more dangerous ultraviolet range. They also spread more easily over the skin. These same nanoparticles are also used in food packaging to reduce UV exposure and prolong shelf life.

9. Many drink bottles are made from plastics containing nanoclays, which increase resistance to permeation by oxygen, carbon dioxide, and moisture. This helps retain carbonation and pressure and increases shelf life by several months.

10. Thanks to nanotechnology, a huge variety of chemical sensors can be programmed to detect a particular chemical at amazingly low levels, for example, a single molecule out of billions. This capability is ideal for surveillance and security systems at labs, industrial sites, and airports. On the medical front, nanosensors can also be used to accurately identify particular cells or substances in the body

intermolecular and intramolecular bonding differwnce ?2 marks

Ans: Intramolecular bonding (chemical interactions): these are bondings that involve changes in the chemical structure of the molecules and include ionic, covalent and metallic bonds;

Intermolecular bonding (physical interaction): these are bondings that do not involve changes in the chemical structure of the molecules and include ion-ion and ion-dipole interactions; van der Waals interactions; hydrogen bonds; hydrophobic interactions; repulsive forces (such as steric repulsions).

quantum confinment 2 marks

Ans: The general principle is that confinement occurs if a characteristic size is less than or equal to the electron coherence length.

Quantum confinement causes the energy of the band gap to increase.

At very small dimensions when the energy levels are quantified, the band overlap present in metals disappears and is actually transformed into a band gap.

This explains why some metals become semiconductors as their size is decreased.

The increase in band gap energy due to quantum confinement means that more energy will be needed in order to be absorbed by the band gap of the material

biomimetic membrane application 3 marks

Ans: the important function of maintaining osmoregulation in living organisms by facilitating water transport through cell walls.

Aquaporin molecule is the ability to restrict the passage of contaminants including bacteria, viruses, minerals, proteins, DNA, dissolved gases, salts, detergents, and even protons without encumbering the passage of water.

For the past several years, a number of research efforts have been underway in an attempt to utilize aquaporins in the development of a new biomimetic membrane technology.

Biomimetic membranes hold considerable promise as a low-energy option and a more effective alternative to desalination, industrial water treatment and other water purification applications.

The Aquaporin Inside technology is essentially a thin film coating that hosts aquaporin proteins in an environment that retains the molecules natural activity of moving water molecules.